
Design and Implementation of a Corporate Company Network Scenario

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Introduction

This corporate network scenario focuses on designing and implementing a Local Area Network (LAN) for a multi-department company consisting of the Marketing, Sales and General Support departments. The goal of the design is to interconnect all departmental workstations, servers and communication systems in a way that ensures data exchange, secure communication and efficient resource sharing across the organization. The proposed network integrates modern networking technologies such as VLAN segmentation, VoIP telephony, dynamic addressing, routing protocols and centralized services. By structuring the network logically and physically, the design ensures that each department operates independently while still maintaining controlled access to shared resources such as email, databases and the company intranet. This project applies structured network design principles to create a scalable, secure and high-performance infrastructure that meets the operational needs of the company and supports future growth.

Objectives

The primary objective of the proposed network is to build an efficient infrastructure that supports the company's functional, user and application requirements. The design aims to enhance communication, improve data accessibility and ensure secure and uninterrupted operations across all departments. The key objectives include:

- 1) Modernize the existing network by introducing a structured topology that supports departmental segmentation through VLANs.
- 2) Enable seamless communication across Marketing, Sales and General Support while maintaining logical separation for security and performance.
- 3) Support VoIP telephony by integrating voice and data over the same network infrastructure without compromising quality.
- 4) Provide centralized services, including email, database hosting and intranet access through dedicated servers in the server room.
- 5) Ensure high availability by designing a network that minimizes downtime and supports continuous user operations.
- 6) Enhance security through the implementation of an ASA firewall, controlled access and traffic filtering.
- 7) Allow scalability so the network can accommodate future expansions in users, devices and services.
- 8) Optimize network performance using appropriate routing protocols (such as RIP), DHCP for dynamic addressing and multilayer switching for efficient traffic handling.

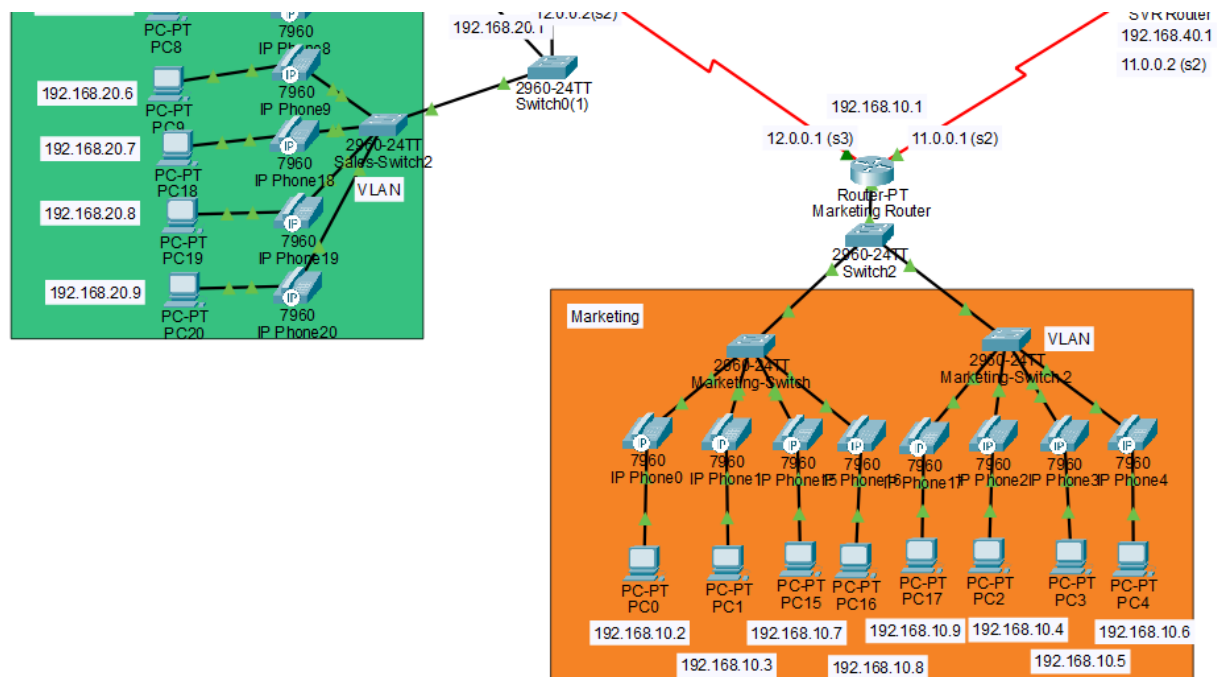
Network Requirements

- 1: The new system should be able to reduce internet downtime. Download and upload links should be maintained above 5 Mbps speed requirement.
- 2: Network will be scalable.
- 3: The system should support remote access.
- 4: Should comprise of data centers with necessary security features and support.

Major Design and Functional Areas

The proposed network uses Cisco multilayer switches and routers to support both data and voice traffic. Departments are divided into separate VLANs, which improves security and reduces broadcast traffic. Voice VLANs are used to prioritize VoIP traffic and maintain call quality.

Routing between VLANs is handled by routing devices configured with RIP, allowing efficient communication across the network. The design follows a hierarchical structure, making management, monitoring, and troubleshooting easier.



Existing Infrastructure

The existing system is quite a basic system. Corporate Company mainly comprises of three main sections as:

- 1) Marketing Department
 - 2) Sales Department
 - 3) General Support Department
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Network Devices

To develop a scalable corporate network, a range of networking devices and components are required. The essential devices include:

- Cisco 2811 Router (R-2811): Provides inter-network communication, supports dynamic routing protocols, and enables connectivity between departmental VLANs and external networks.
 - Cisco 3560-24PS Multilayer Switch (SW-3560): Handles Layer 2 and Layer 3 switching, supports VLAN segmentation, inter-VLAN routing, QoS, and ACLs for traffic control.
 - Cisco IP Phone 7960 (IP-7960): Enables VoIP communication by transmitting voice over the data network using dedicated voice VLANs.
 - HTTP / Email / FTP Servers
 - Generic Workstations (PC-MKT, PC-SLS, PC-SUP): Employee computers used for daily operations, connected to departmental VLANs.
 - Network Interface Cards (NIC-G1): Provide Gigabit Ethernet connectivity for all PCs and servers.
 - Category 5E STP Cabling (CAB-CAT5E): Ensures stable wired connections for both data and VoIP traffic across all floors.
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Request for Proposal (RFP)

Serial no	Network Devices	Price per unit (Approx.)	Quantity	Total cost (EGP)
1	Cisco multilayer switch	18,000	1	18,000
2	Cisco Router	12,000	4	48,000
3	Wires/Cables	2,500 per bundle	10	25,000
4	Cisco IP Phone	1,500	75	112,500
5	HTTP/FTP/EMAIL Servers	35,000 each	3	105,000
6	PCS	8,000	75	600,000
			Total	908,500

IP Addressing Plan

Marketing Department (192.168.10.0)

PC-0	192.168.10.2
PC-1	192.168.10.3
PC-2	192.168.10.4
PC-3	192.168.10.5
PC-4	192.168.10.6
PC-15	192.168.10.7
PC-16	192.168.10.8
PC-17	192.168.10.9

Sales Department (192.168.20.0)

PC-5	192.168.20.2
PC-6	192.168.20.3
PC-7	192.168.20.4
PC-8	192.168.20.5
PC-9	192.168.20.6
PC-18	192.168.20.7
PC-19	192.168.20.8
PC-20	192.168.20.9

General Support Department (192.168.30.0)

PC-10	192.168.30.2
PC-11	192.168.30.3
PC-12	192.168.30.4
PC-13	192.168.30.5
PC-14	192.168.30.6

Server Room (192.168.40.0)

Server-PT Email	192.168.40.2
Server-PT FTP	192.168.40.3
Server-PT HTTP	192.168.40.4

Routing Protocol Plan

Routing Information Protocol (RIP) is a dynamic routing protocol which uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance vector routing protocol which has AD value 120 and works on the application layer of OSI model.

The image displays four screenshots of network router configuration interfaces, each showing the configuration of the Routing Information Protocol (RIP) on a different router: SVR Router, Support Router, Sales-Router, and Marketing Router. Each interface is divided into several sections:

- Physical:** Shows the router's physical attributes.
- Config:** The main configuration area, further divided into:
 - GLOBAL:** Contains settings for the router.
 - ROUTING:** Contains settings for the routing protocol.
 - INTERFACE:** Contains settings for the router's interfaces.
- Attributes:** Shows the router's attributes.

In the **ROUTING** section, the **RIP** protocol is selected. The **INTERFACE** section shows the configuration for the **FastEthernet0/0** interface. The **Network** section shows the configuration for the **192.168.10.0** network. The **Equivalent IOS Commands** section shows the commands used to configure the router:

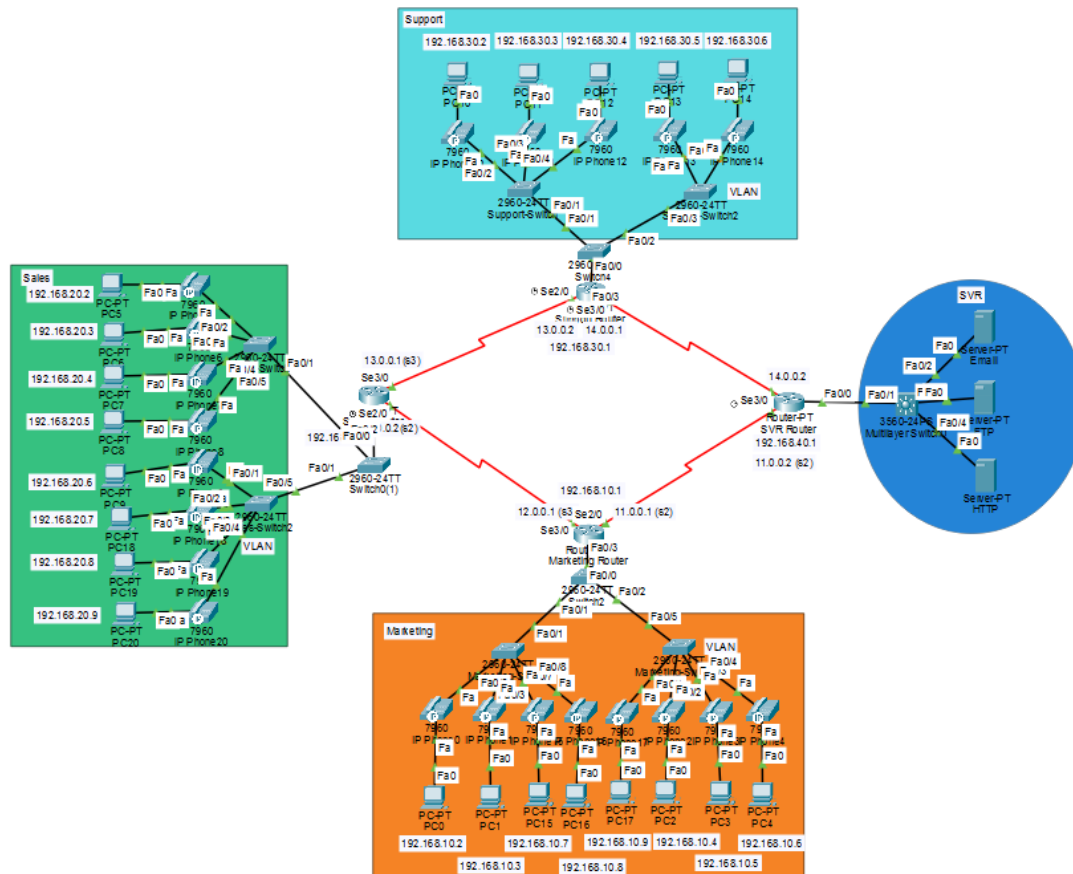
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%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

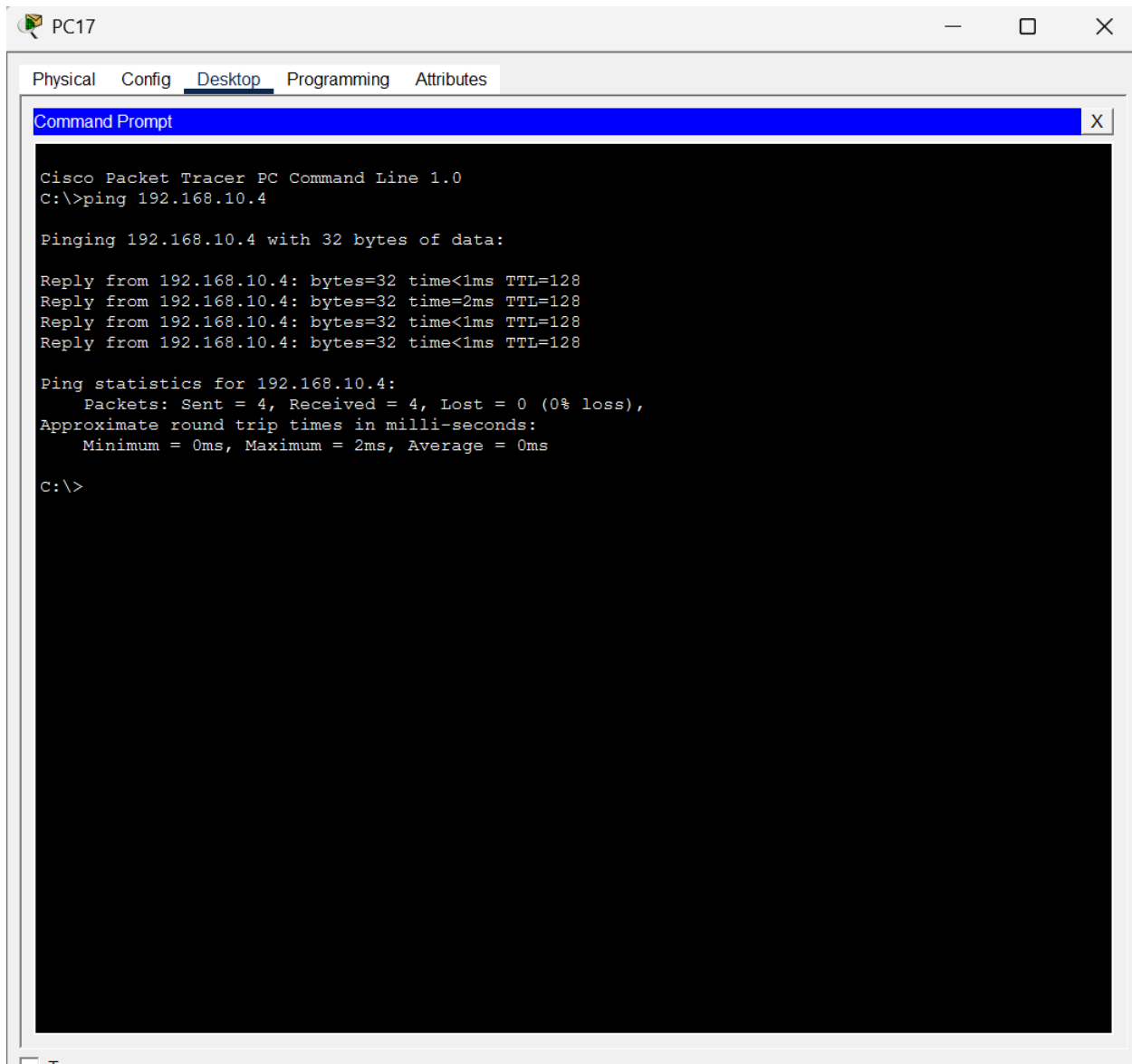
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
  
```

The **Support Router** and **Marketing Router** screenshots show the same configuration steps, but with the **Serial2/0** interface selected in the **INTERFACE** section. The **Sales-Router** screenshot shows the **Serial3/0** interface selected in the **INTERFACE** section.

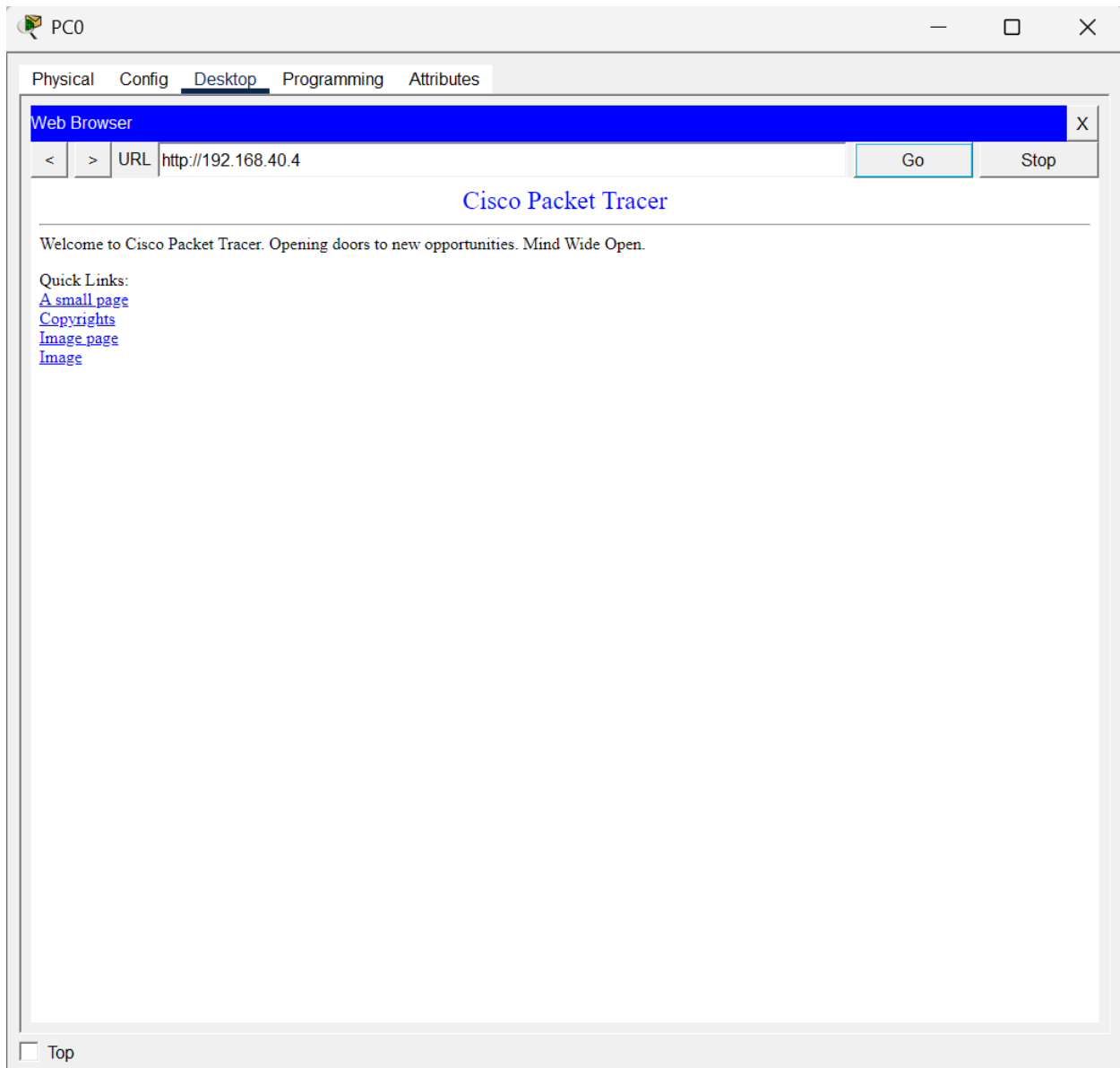
Network Design and Implementation



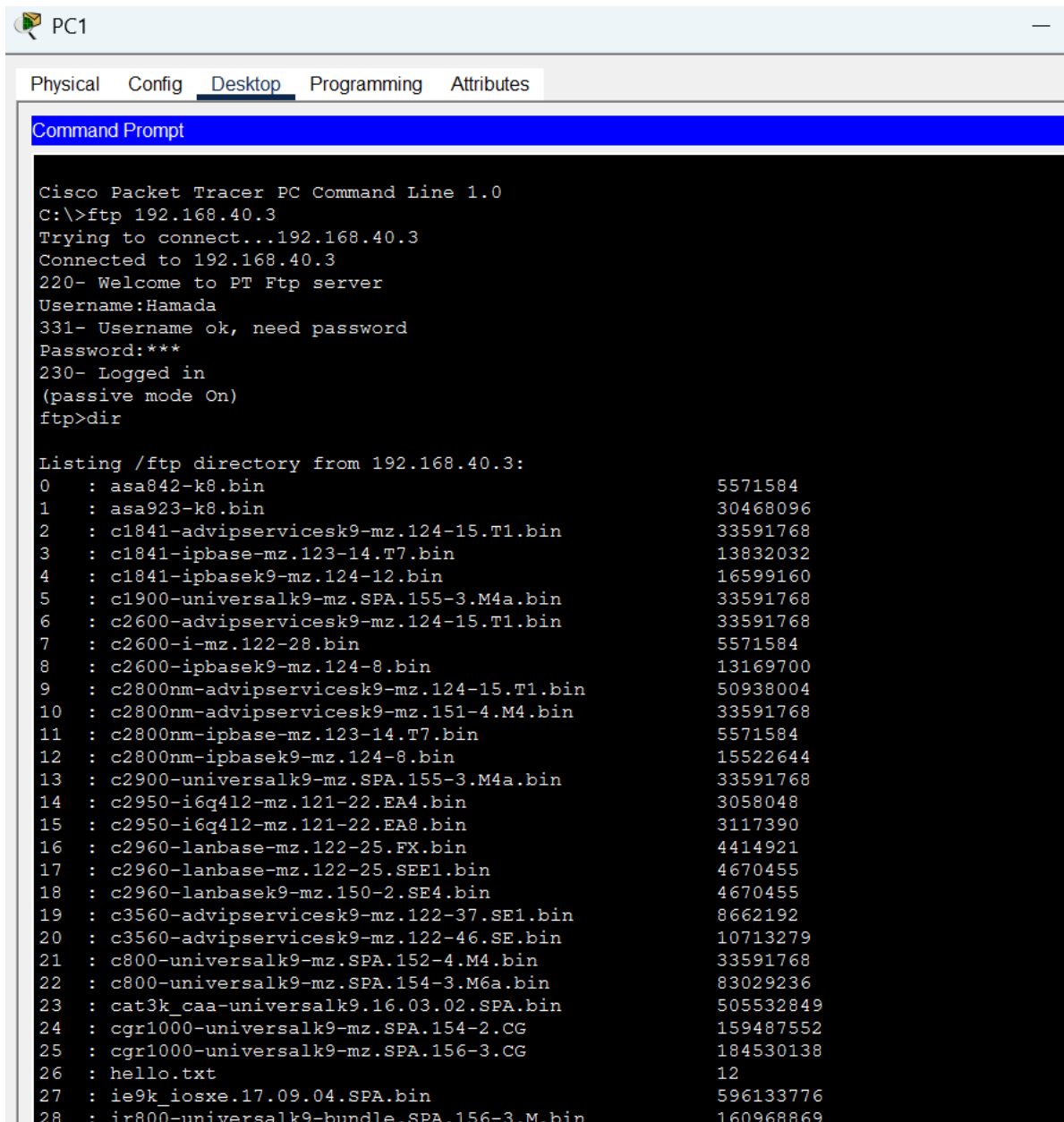
The prototype of the proposed network is implemented on cisco packet tracer



Testing VLAN communications



Testing Web Hostin



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.40.3
Trying to connect...192.168.40.3
Connected to 192.168.40.3
220- Welcome to PT Ftp server
Username:Hamada
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.40.3:
0   : asa842-k8.bin                5571584
1   : asa923-k8.bin                30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin  33591768
3   : c1841-ipbase-mz.123-14.T7.bin  13832032
4   : c1841-ipbasek9-mz.124-12.bin  16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin  33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
7   : c2600-i-mz.122-28.bin        5571584
8   : c2600-ipbasek9-mz.124-8.bin  13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin  15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin  3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin  3117390
16  : c2960-lanbase-mz.122-25.FX.bin  4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin  4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin  4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin  8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG  159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG  184530138
26  : hello.txt                   12
27  : ie9k_iosxe.17.09.04.SPA.bin  596133776
28  : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869

```

Testing FTP Server

CLI

```
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool VOICE20
Router(dhcp-config)#EX
Router(config)#tele
      ^
% Invalid input detected at '^' marker.

Router(config)#telephony-service
      ^
% Invalid input detected at '^' marker.

Router(config)#telephony-service
      ^
% Invalid input detected at '^' marker.

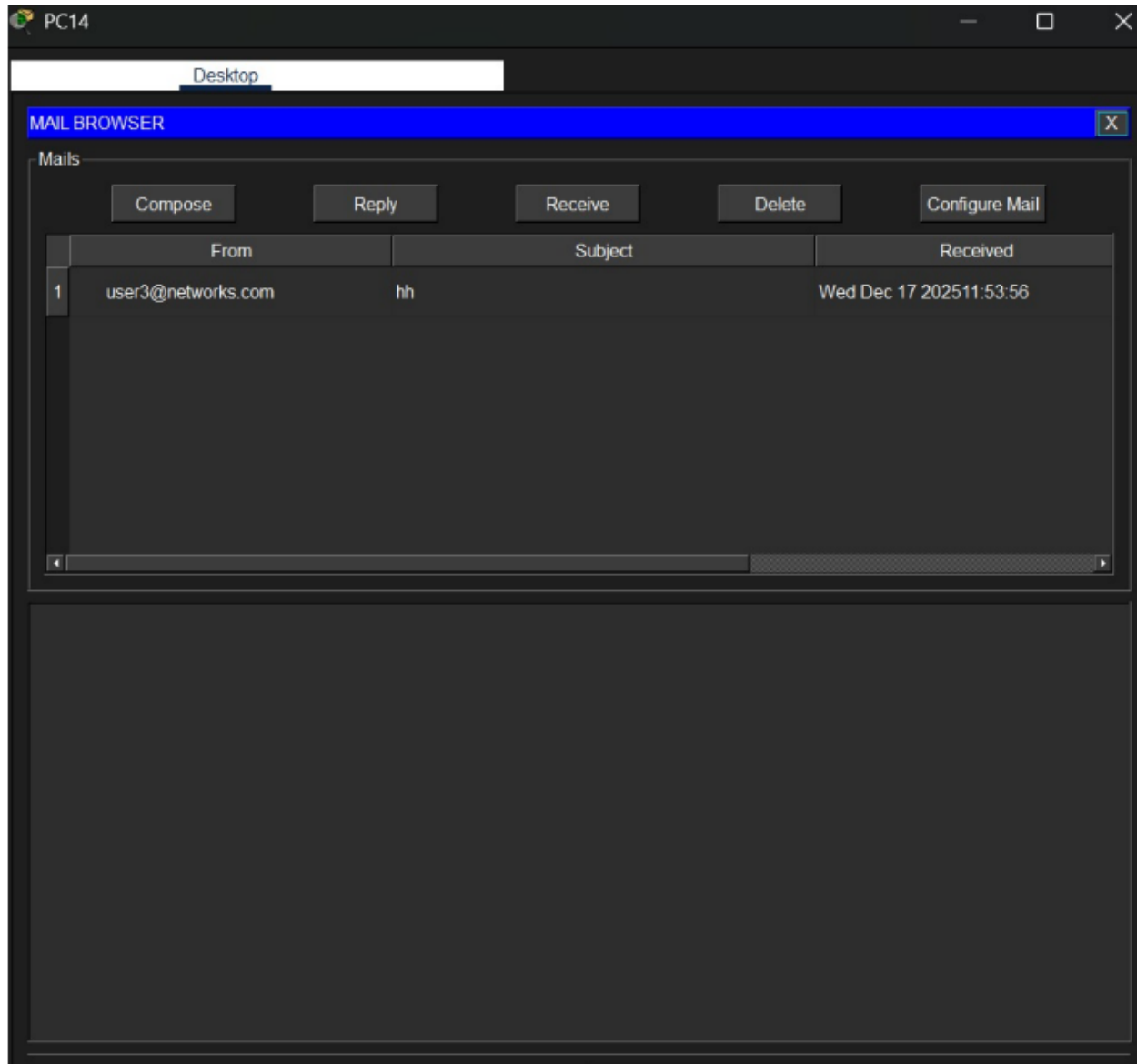
Router(config)#telephony-service
      ^
% Invalid input detected at '^' marker.

Router(config)#
```

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Testing IP phones



Testing Email